VRP variants for quantum optimization

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1 VRP

1.1 Review

Existing formulations on VRP and its variants are summarized in Table 1. There are some formulations in [4], however I think they are either wrong or focusing on another problem.

From [15]:

"The two-index formulations VRP1 and VRP2 model the underlying fleet only implicitly. Even if we know that a specific edge i, j or arc (i, j) is used in a solution, it is not clear which vehicle k will travel between i and j. Hence, two-index formulations cannot model vehicle-specific characteristics such as different capacities, associated depots, and costs (see Section 1.3.4), which is a clear disadvantage"

1.2 Our model

We assume we have K vehicles which need to run over $V = \{0, 1, \dots, N\}$ cities. The cost of moving from city $v \in V$ to $w \in V \setminus \{v\}$ equals $d_{v,w}$ we want to

Problem	Year	Assumptions / Notes	Number of variables	Ref.	Type
VRP	2020		$\mathcal{O}(KN^2)$	[6]	QUBO
VRP	2022		$\mathcal{O}(KN^2)$	[8]	QUBO
VRP	2022	We think this is wrong because it omits the subtour conditions	$\mathcal{O}(N^2)$	[5]	QUBO
VRP		This is MTZ formulation, kind of the standard one apppearing in multiple places	$\mathcal{O}(N^2)$	[1]	ILP/QUBO
VRP	2020	They have a fractional objective, why this is working is not clear	$\mathcal{O}(KN^2)$	[2]	ILP
CVRP	1975		$\mathcal{O}(N^3)$	[7]	ILP
CVRP	2008		$\mathcal{O}(KN^2)$	[10]	QP
CVRP	2003	Identical capacities	$\mathcal{O}(N^2 + N\log\overline{T}_k) = \mathcal{O}(N^2)$	[11]	ILP
CVRP	2010	Identical capacities, extension to MTZ	$\mathcal{O}(N^2 + N\log\overline{T}_k) = \mathcal{O}(N^2)$	[12]	ILP
CVRP	2016	Identical capacities, they say they use cts variables but I dont see the need	$\mathcal{O}(N^2 + N\log\overline{T}_k) = \mathcal{O}(N^2)$	[14]	ILP
VRPTW	2020		$\mathcal{O}(KN^3)$	[9]	ILP/QUBO
VRPTW	2020		$\mathcal{O}(KN^2)$	[9]	ILP/QUBO
VRPTW	1997		$\mathcal{O}(KN^2 + K\log\bar{l}_v) = \mathcal{O}(KN^2)$	[13]	ILP
CVRPTW	2016	Identical capacities, they say they use cts variables but I dont see the need	$\mathcal{O}(N^2 + N\log\overline{T}_k + K\log\bar{l}_v) = \mathcal{O}(N^2)$	[14]	ILP
CVRPTW	2017	Identical capacities	$\mathcal{O}(KN^2 + K\log\bar{l}_v) = \mathcal{O}(KN^2)$	[3]	ILP

Table 1: Literature on VRP and its variants.

minimize the total cost of moving vehicles. Let N be the depot, in which all vehicles starts and ends.

Let us consider $V' = V \cup \{N+1, \dots, N+K-1\}$, where $N, \dots, N+K-1$ are copies of the depot. For $v, w \in \{N, \dots, N+K-1\}$, we define $d_{v,w} = 0$. Now, when the problem of VRP is relaxed to the case where we have single route passing through all V', the routes should be splitted and the i-th route should be attached to the i-th vehicle.

Let $x_{t,v}$ be binary variable that is 1 if city $v \in V'$ is visited at step $t = 0, \dots, |V'|-1$. Then, the VRP takes the form

$$\begin{aligned}
\min \ & \sum_{t=0}^{|V'|-1} \sum_{\substack{v\in V',\ w\in V' \\ v\neq w}} d_{v,w} x_{t,v} x_{t+1,w} \\
\text{s.t.} \ & \sum_{t=0}^{|V'|-1} x_{t,v} = 1 && v \in V' \\
& \sum_{v\in V'} x_{t,v} = 1 && t = 0, \dots, |V'|-1 \\
& \sum_{v=N}^{N+K-1} (x_{t,v} + x_{t+1,v}) \leq 1 && t = 0, \dots, |V'|-1
\end{aligned}$$

We assume that $|V'| \to 0$ simplification for the indices. The first two conditions ensure that $x_{t,v}$ forms a permutation, which is equivalent to route passing K times through the depot. The last condition certifies that we can't go from depot to depot, which in essence forces having K vehicles exactly. In case we omit this condition, we would have at most K vehicles.

The formulation above requires $(N+K-1)^2$ binary variables $x_{t,v}$ and $N+K-1$ bits for making the last condition an equality constraint (which is necessary step to transform it to QUBO). In total one need $(N+K-1)(N+K) = \mathcal{O}(N^2)$, where we used the fact $K \leq N$. Alternatively, instead of incorporating last inequality by first transforming it to an equality, one can add the terms $x_{t,v}x_{t+1,v}$ directly to the objective function, without the need for slack variables.

Note that we can slightly simplify the above QIP by setting $x_{0,N} = 1$, and thus $x_{0,v} = x_{t,N} = 0$ for $v \in V'/\{N\}$ and $t > 0$. This has negligible impact on the qubits cost.

2 CVRP

Here, we assume every vehicle $k \in \{0, \dots, K-1\}$ has some capacity T_k of goods it can handle, and each city $v \in V \setminus \{N\}$ (except the depot) has a weight $w_v > 0$ which says how much goods it would like to receive. We assume $w_N = 0$

From know we consider equivalent model in which the vehicle departures depot with no goods and it picks up them from the cities it visits. We will use the same concept of multiplying the depots as before. We introduce two classes of integer variables: $0 \leq L_t \leq \sum_{v \in V \setminus \{N\}} w_v$ which says how many goods vehicle has while leaving the city it visited at time t; $1 \leq C_t \leq K$ which says which vehicle is used at time t.

We will assure the following conditions: if city at time t is a depot then $L_t = 0$, and if it is city v which is not a depot we have $L_{t+1} = L_t + w_v$. We assume we always start at the depot at time $t = 0$ and therefore $C_0 = 0$, and if at time $t > 0$ the city visited is not a depot we have $C_{t+1} = C_t$, and otherwise

$C_{t+1} = C_t + 1$. The following formulation describes the CVRP problem

$$
\begin{aligned}
\min \; & \sum_{t=0}^{|V'|-1} \sum_{\substack{v \in V',\, w \in V' \\ v \neq w}} d_{v,w} x_{t,v} x_{t+1,w} & \\
\text{s.t.} \; & \sum_{t=0}^{|V'|-1} x_{t,v} = 1 & v \in V' \\
& \sum_{v \in V'} x_{t,v} = 1 & t = 0, \dots, |V'| - 1 \\
& \sum_{v=N}^{N+K-1} (x_{t,v} + x_{t+1,v}) \leq 1 & t = 0, \dots, |V'| - 1 \\
& x_{0,N} = 1 & \\
& \sum_{v=N}^{N+K-1} x_{t,v} = 1 \implies L_t = 0 & t = 0, \dots, |V'| - 1 \\
& \sum_{v=N}^{N+K-1} x_{t+1,v} = 0 \implies L_{t+1} = L_t + \sum_{v=0}^{N-1} w_v x_{t+1,v} & t = 0, \dots, |V'| - 2 \\
& C_0 = 0 & \\
& C_{t+1} = C_t + \sum_{v=N}^{N+K-1} x_{t,v} & t = 0, \dots, |V'| - 2 \\
& C_t = k \implies L_t \leq T_k & t = 0, \dots, |V' - 1|; k = 1, \dots, K
\end{aligned}
$$

We will need $(N+K-1)^2$ bits for $x_{t,v}$ variables, $(N+K-1)\log((N+K-1)\max_v w_v)$ bits for L_t variables, $(N+K-1)K$ bits for C_t variables (they are encoded 1-hot so that we have variables of the form $C_{t,1}, C_{t,2}, \dots, C_{t,K}$) and $N+K-1$ bits for transforming the inequality.

For transforming conditionals to equalities, for the first and second one, we need $3(N+K-1)\log((N+K-1)\max_v w_v)$ bits in total and $((N+K-1)K \log \max_k T_k)$ bits for transforming the last one.

Overall, assuming w_v, T_k grows at most polynomially with N we need $\mathcal{O}\big((N+K-1)^2 \log N\big) = \mathcal{O}\big(N^2 \log N\big)$ bits.

2.1 Linearization of the conditionals

Notation: $\overline{L}_t = \max L_t \leq \sum_{v \in V \setminus \{N\}} w_v$

Cond 1.

$$\sum_{v=N}^{N+K-1} x_{t,v} = 1 \implies L_t = 0 \qquad t = 0, \dots, |V'| - 1 \tag{1}$$

$$L_t \leq \overline{L}_t (1 - \sum_{v=N}^{N+K-1} x_{t,v}) \tag{2}$$

Cond 2.
We can select $\mu = \overline{L}_t$.

$$\sum_{v=N}^{N+K-1} x_{t,v} = 0 \implies L_{t+1} = L_t + \sum_{v=0}^{N-1} w_v x_{t+1,v} \qquad t = 0, \dots, |V'| - 2 \tag{3}$$

$$L_{t+1} - L_t - \sum_{v=0}^{N-1} w_v x_{t+1,v} \leq \mu \sum_{v=N}^{N+K-1} x_{t,v} \tag{4}$$

$$L_{t+1} - L_t - \sum_{v=0}^{N-1} w_v x_{t+1,v} \geq -\mu \sum_{v=N}^{N+K-1} x_{t,v} \tag{5}$$

Cond. 3

$$C_t = k \implies L_t \leq T_k \qquad t = 0, \dots, |V' - 1|; k = 1, \dots, K \tag{6}$$

$$L_t \leq \sum_k T_k C_{t,k} \tag{7}$$

Note: Let $\omega_0, \omega_1, \dots, \omega_{N-1}$ be the list of ordered weights such that $\omega_i \geq \omega_{i+1}$ for $i = 0, \dots, N-2$. Can we use a more strict bound: $L_t \leq \sum_{i=0}^{t} \omega_i$?

3 VRPTW

In here we assume each city $v \in \{0, 1, \dots, N-1\}$ needs to be visited within time slot $[e_v, l_v]$ where both e_v, l_v are positive integer numbers. We assume $e_N, l_N = 0$.

Let $x_{t,v}$ be as before, $0 \leq A_t \leq \max_{v \in V \setminus \{N\}} l_v$ be arrival time to the city visited at time t, and W_t is a waiting time, which is time spend in city visited at time t. Let $\tau_{v,w}$ be the time required for moving from v to w. The model takes the form

$$
\begin{aligned}
\min \; & \sum_{t=0}^{|V'|-1} \sum_{\substack{v\in V', w\in V'\\ v\neq w}} d_{v,w} x_{t,v} x_{t+1,w} & \\
\text{s.t.} \; & \sum_{t=0}^{|V'|-1} x_{t,v} = 1 & v \in V' \\
& \sum_{v\in V'} x_{t,v} = 1 & t = 0,\dots,|V'|-1 \\
& \sum_{v=N}^{|V'|-1} (x_{t,v} + x_{t+1,v}) \leq 1 & t = 0,\dots,|V'|-1 \\
& x_{0,N} = 1 & \\
& \sum_{v=0}^{N-1} x_{t,v} = 1 \implies A_{t+1} = A_t + W_t + \sum_{w\in V'\setminus\{v\}} \tau_{v,w} x_{t+1,w} & t = 0,\dots,|V'|-2 \\
& \sum_{v=0}^{N-1} e_v x_{t,v} \leq A_t \leq \sum_{v=0}^{N-1} l_v x_{t,v} & t = 0,\dots,|V'|-1
\end{aligned}
$$

We need $(N+K-1)^2$ qubits for $x_{t,v}$ variables, $(N+K-1)\log\max_v l_v$ for each A_t and W_t variables and $(N+K-1)+2(N+K-1)\log\max_v l_v$ slack variables for transforming the inequalities.

To transform the conditional, we need $2(N+K-2)\log\max_{v,w}\tau_{v,w}$ variables. Note that $\max_{v,w}\tau_{v,w}$ is upper bounded by $\max_v l_v$

Assuming l_v grows at most polynomially with N we need $\mathcal{O}\big((N+K-1)^2\log N\big) = \mathcal{O}\big(N^2\log N\big)$ variables.

3.1 Linearization of the conditional

$$\sum_{v=0}^{N-1} x_{t,v} = 1 \implies A_{t+1} = A_t + W_t + \sum_{w\in V'\setminus\{v\}} \tau_{v,w} x_{t+1,w} \qquad t = 0,\dots,|V'|-2 \tag{8}$$

$$A_{t+1} - A_t - W_t - \sum_{w\in V'\setminus\{v\}} \tau_{v,w} x_{t+1,w} \geq -\mu(1 - \sum_{v=0}^{N-1} x_{t,v}) \tag{9}$$

$$A_{t+1} - A_t - W_t - \sum_{w\in V'\setminus\{v\}} \tau_{v,w} x_{t+1,w} \leq \mu(1 - \sum_{v=0}^{N-1} x_{t,v}) \tag{10}$$

We can let $\mu = \max_{v,w}\tau_{v,w}$. Let's see why the choice of μ is valid. Note that $A_{t+1} \geq A_t + W_t$ in general. Since $\sum_{w\in V'\setminus\{v\}} \tau_{v,w} x_{t+1,w} \leq \max_{v,w}\tau_{v,w}$, Eq. (9) is always True. For Eq. (10), note that, $A_{t+1} \leq A_t + W_t + \max_{v,w}\tau_{v,w}$ should be true in general.

4 CVRPTW

Just merge CVRP and VRPTW. Number of qubits is again $\mathcal{O}\big(N^2 \log N\big)$.

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